
ASUS
Bambu Lab
Bambu Lab A1
57°C
100 %
0 %
Purge old filament

Massachusetts Institute of Technology

FDM Variation in Bearing Hole Diameters

2.830 – Control of Manufacturing Processes

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12/09/2025

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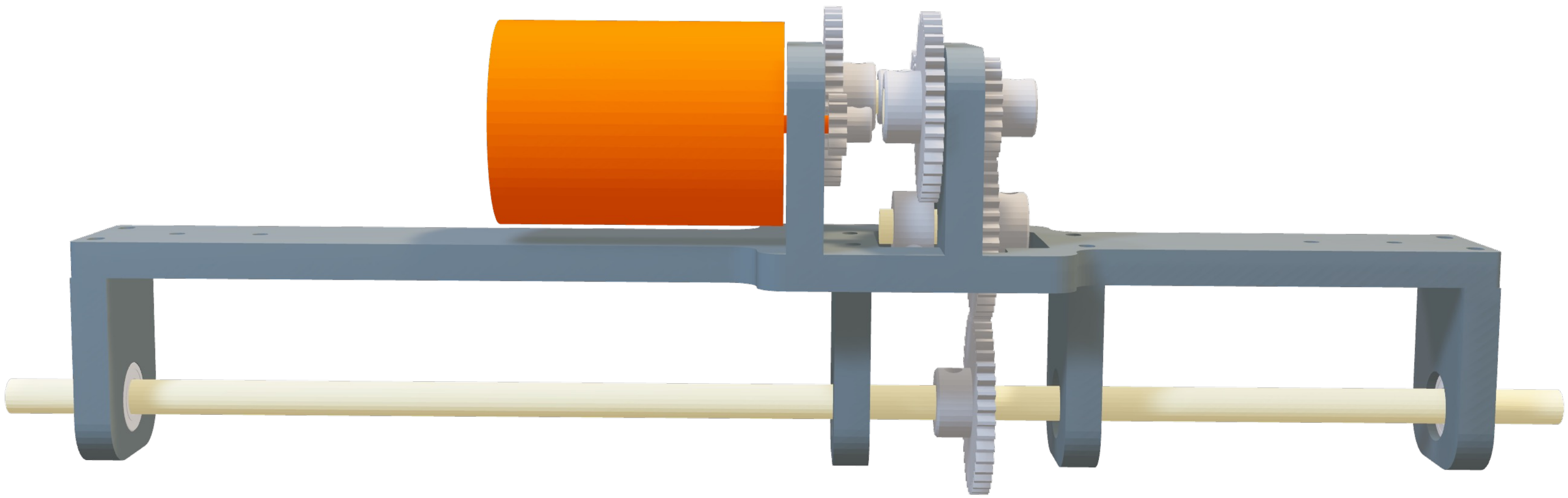
Maximize quality while minimizing cycle times and rework



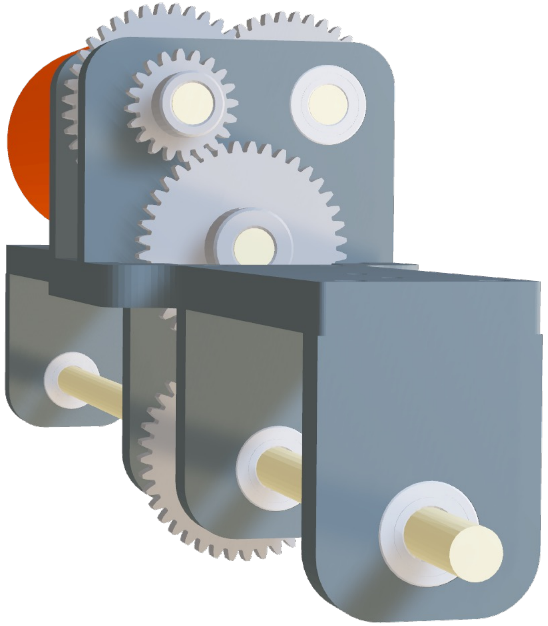
Premise

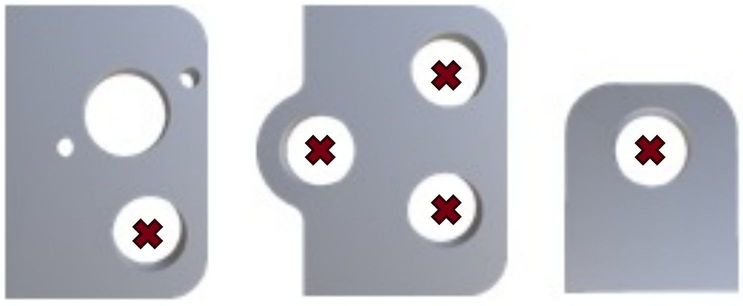
- **Design & Manufacturing of 40 RC cars** as part of a scaled production project
- Mirrors real-world subassembly manufacturing
- We are on the **drivetrain subassembly team** (axle holders, gear & motor mounts)

Goal: Deliver <u>**high-quality**</u> bearing hole parts efficiently by **minimizing printing time and reducing non-conformance issues**.

Problem Identification: Drivetrain Manufacturing

Maximize quality while minimizing cycle times and rework





Bearing hole areas are marked

Prevent rod deflections by holding axle in place

Problem Definition

Data from spec sheets

- Bambu A1 typically has a shrinking size of about 0.1 ~ 0.2 mm, with a 5% variance.
- $10 Amazon bearings has a tolerance of 0.0914 mm
- Desired press fit interference of 0.04 mm

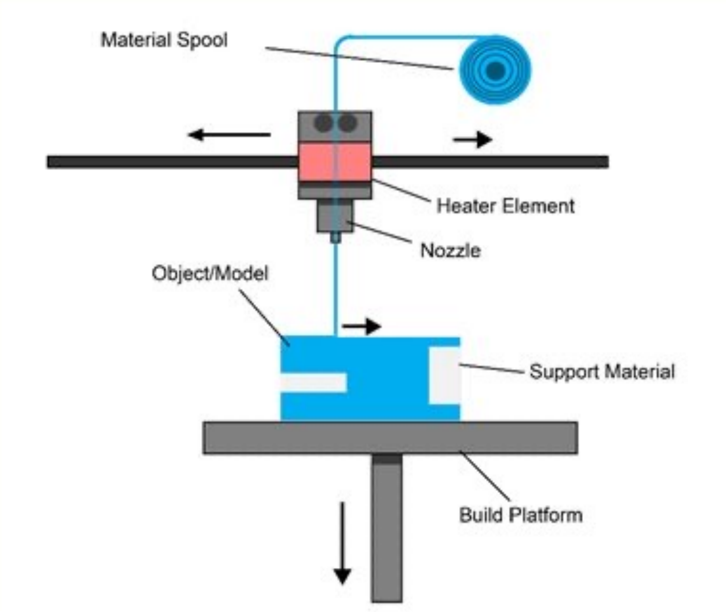
FDM variation causes some axle holders to vary in size, causing deformation of bearing during press-fit.
We need to fit 7 bearing holes per RC car, which gives us 280 press fitted bearing holes in total.

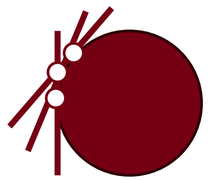
Introduction/Background

Objective: What Parameters are we changing to **increase** efficiency and **reduce** printing time

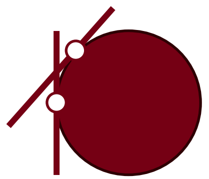
Hypothesis

Printing Speed	↓	↑	Printing Time
Resolution	↓	↓	Variance
Change Filament PLA to PETG		↓	Shrinkage
Airflow	↓	↓	Variance





Low Resolution



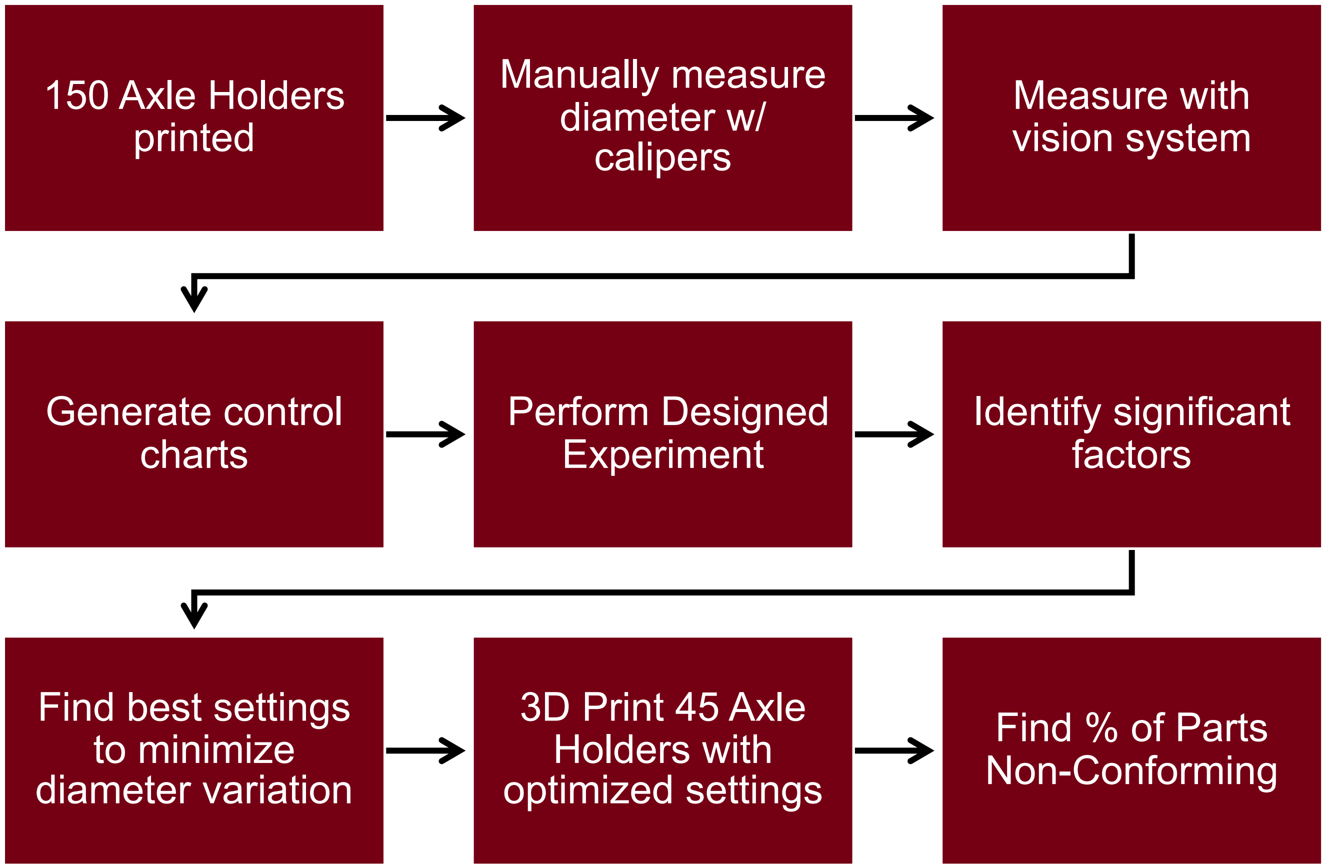
High Resolution

Experimental Methodology

To determine how the data changes over time, we will be using **Control Charts** to identify if the process is out of control which helps with quality control and improvement.

The **process capability metrics** C_p and C_{pk} will be calculated to determine if the process has low spread and is center about the **USL** and **LSL.**

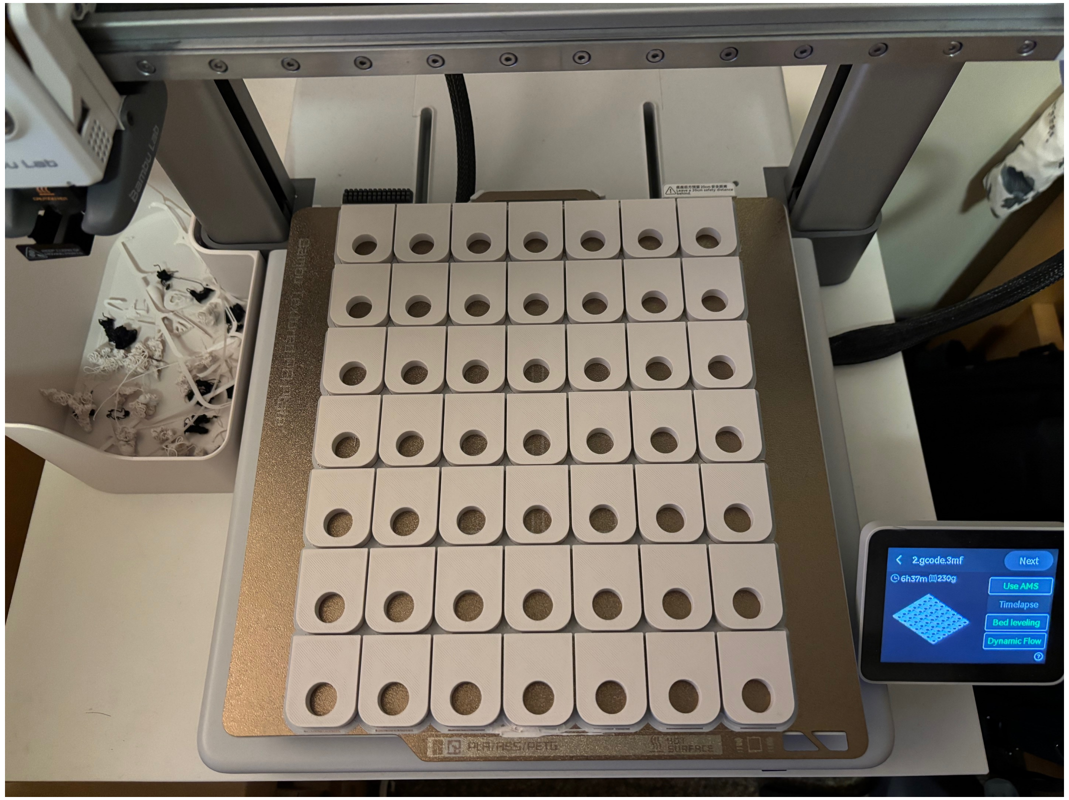
Interpreting this data, we can then establish a **fractional factorial experiment** to create a linear model that will determine the optimal print setting and environment.



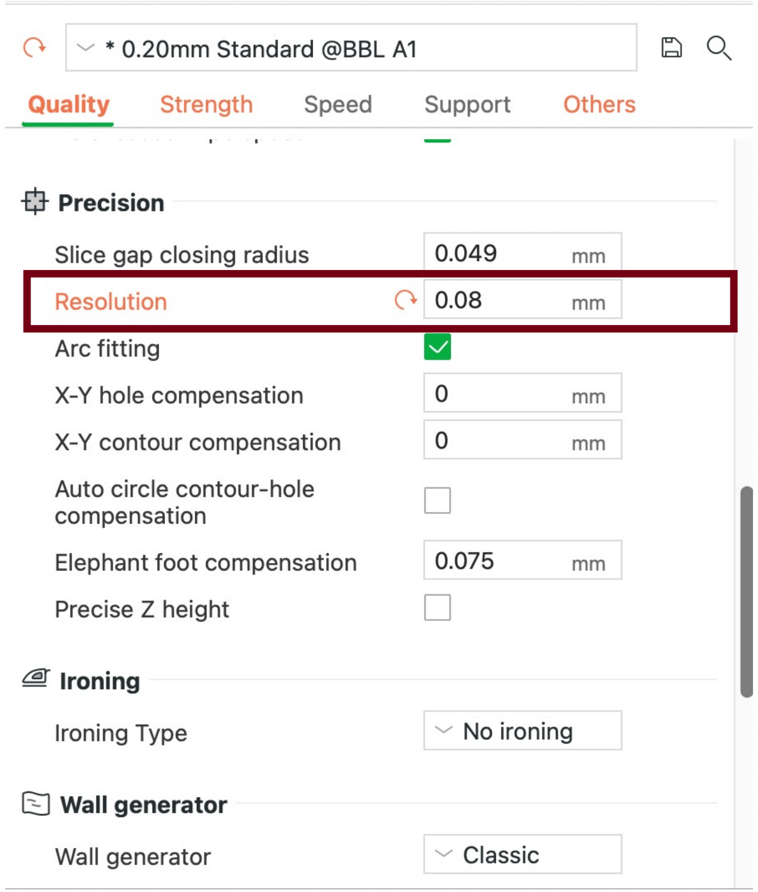
Control Chart

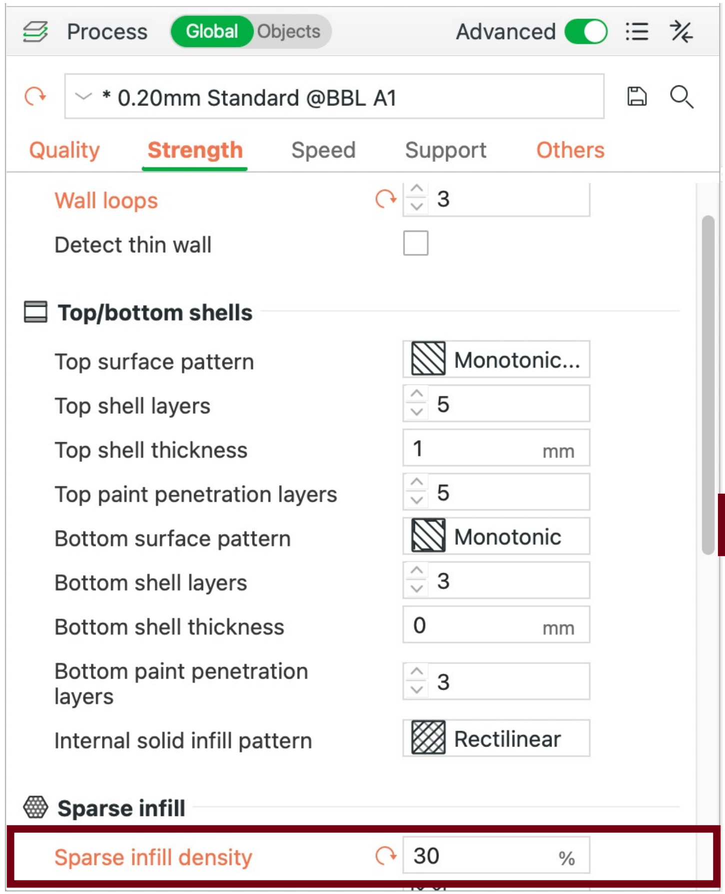
Base parameters were determined based off 3D Printer settings (used a Bambu A1)

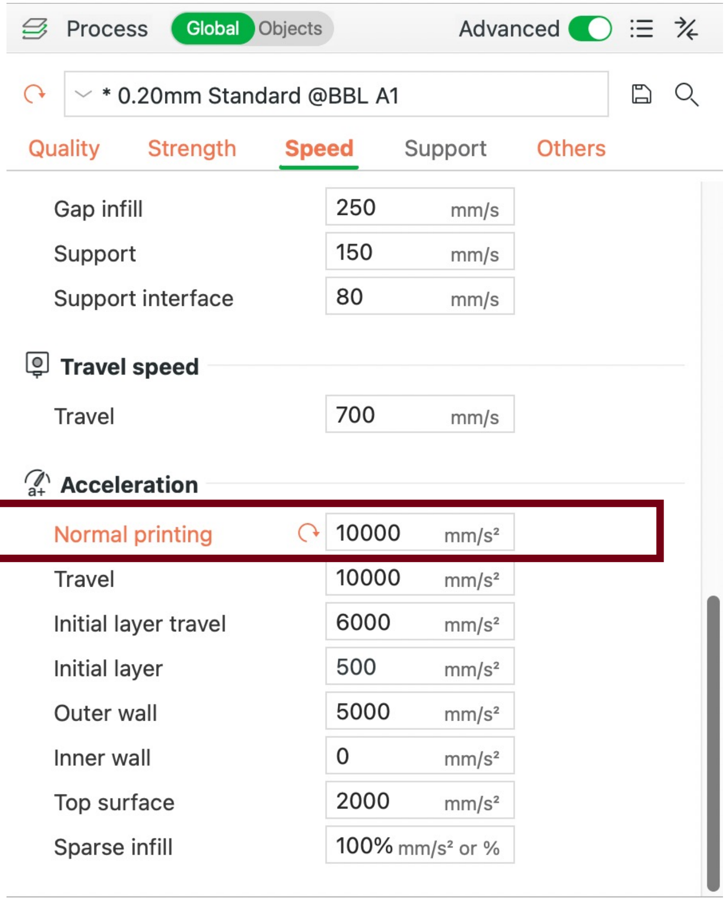
Parameter	Value
Wall Loops	3
Sparse Infill Density	30%
Sparse Infill Pattern	Grid
Nozzle Diameter	0.4 mm
Filament	PLA Matte
Build Plate	Textured PEI Plate
Speed	105 mm/s
Brim Type	Outer brim only
Brim Width	3 mm



BambuStudio Slicer Setting







Part Manufacturing

Manufacturing Batches
- 3 batches produced, 50 parts per batch
- Batch 1 (50 parts) used as phase 1 "historical" data to generate UCL and LCL
- Batch 2 & 3 (100 parts) used as "monitoring" data to detect any mean shifts

Batch 1

Batch 2

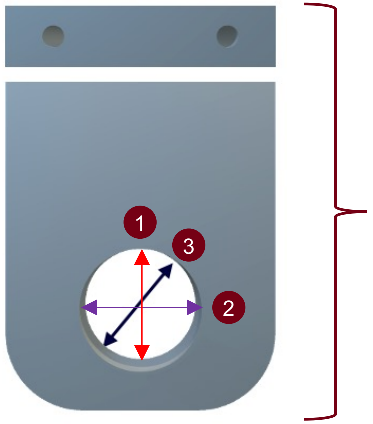
Batch 3

- Phase 1 "Historical" Data
  - 50 parts

- Phase 2 "Monitoring" Data
  - 100 parts

Data Collection

- Each 3D printed part was measured with calipers in three separate locations ten averaged
- Each part was then measured with the Vision System



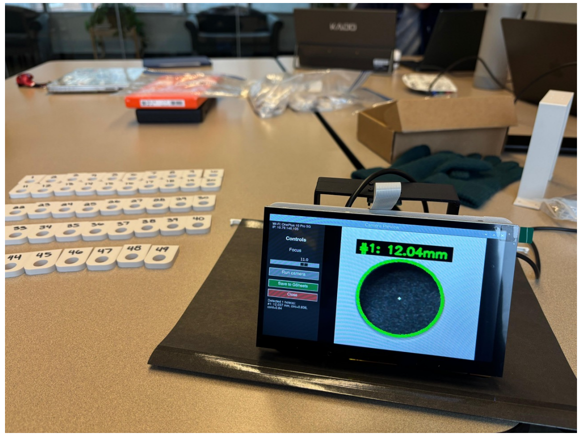
Each diameter was measured in three separate locations, then the average was taken

Number	Digital Measurement	Caliper Measurement 1	Caliper Measurement 2	Caliper Measurement 3	Caliper Measurement Average
1	12.10	12.6	12.73	12.72	12.68
2	12.13	12.62	12.75	12.68	12.68
3	12.10	12.56	12.73	12.68	12.66
4	12.12	12.6	12.73	12.68	12.67
5	12.10	12.57	12.72	12.66	12.65
6	12.06	12.57	12.64	12.67	12.63
7	12.09	12.6	12.56	12.58	12.58
8	12.01	12.58	12.57	12.67	12.61
9	12.06	12.68	12.7	12.66	12.68
10	12.05	12.59	12.69	12.64	12.64

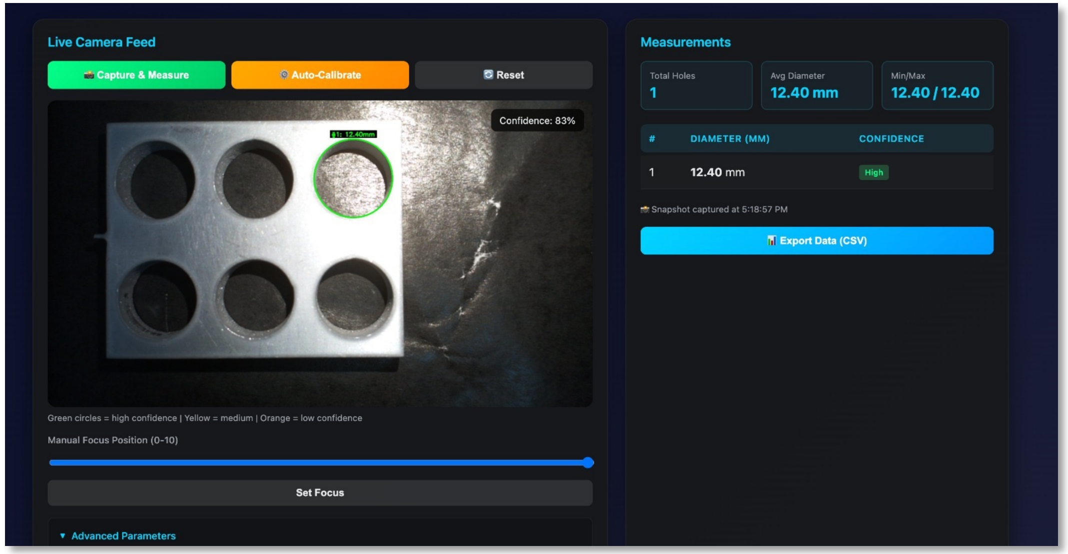
Measuring Methodology - Vision System

Quality Control System for Bearing Press Fit

Raspberry Pi
Camera Module
Set-Up



Web App
Interface



Vision System Setup

- Used Raspberry Pi camera + OpenCV for hole measurement
- Parts placed under camera, multiple measurements taken to estimate hole diameter
- **Data automatically logged** to process measurement file and **run charts**

Current State

- Basic OpenCV measurement pipeline implemented
- System detects and measures target features but is influenced by lighting and positioning parameters

Challenges & Learnings

- Emphasized need for data validation and iterative tuning
- Continued improvements required for repeatability and accuracy

Identify UCL and LCL

- Using Phase 1 data as the base set for the data determine mean and range for each subgroup]
  - **Note: We use range instead of standard deviation since n is small**

$$UCL = \bar{\bar{x}} + 3\frac{\bar{R}}{d_2\sqrt{n}} \qquad LCL = \bar{\bar{x}} - 3\frac{\bar{R}}{d_2\sqrt{n}}$$

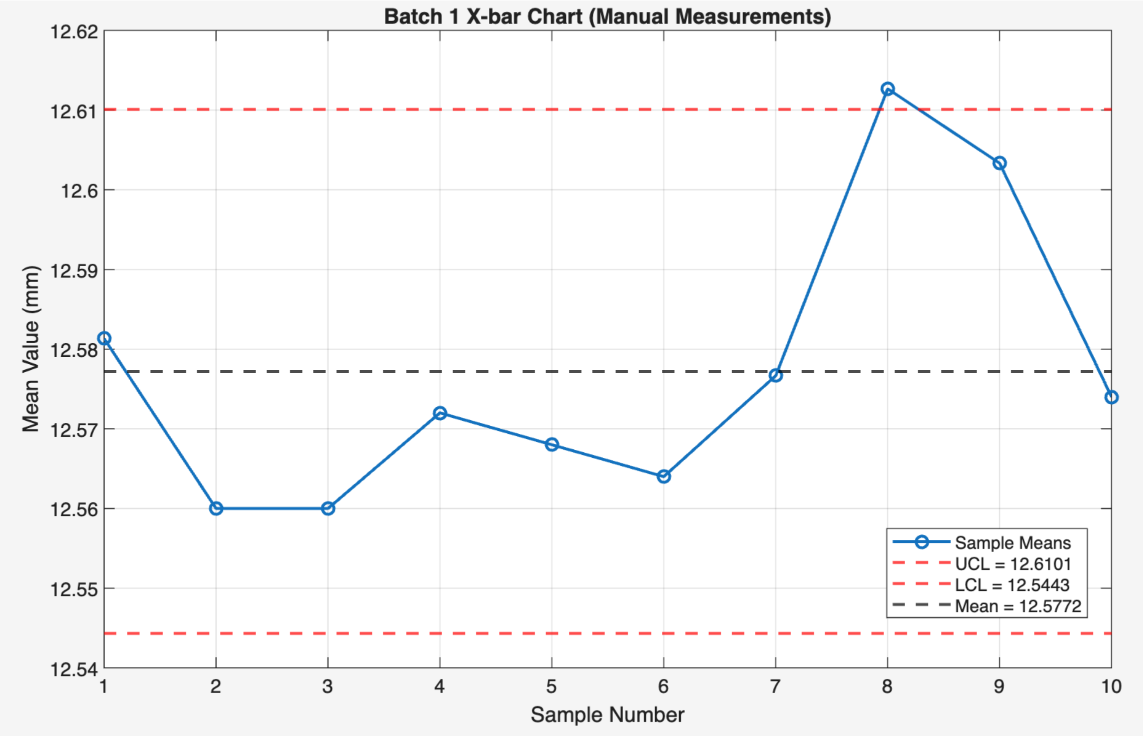
$$where\ \bar{\bar{x}} = 12.5772 \quad d_2 = 2.326 \quad n = 5$$

Calculated **UCL = 12.61** and **LCL = 12.544**

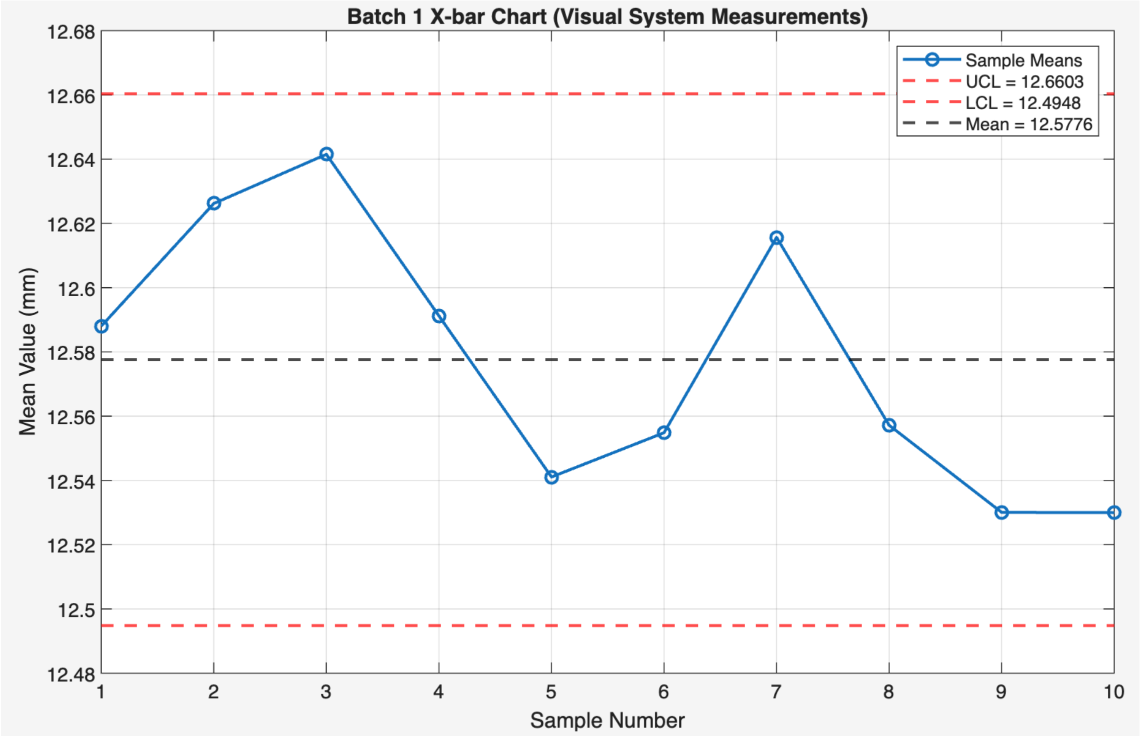
Subgroup	Group Mean	Range
1	12.58	0.06
2	12.56	0.043
3	12.56	0.11
4	12.57	0.036
5	12.57	0.073
6	12.56	0.083
7	12.58	0.033
8	12.61	0.05
9	12.60	0.033
10	12.57	0.05

Batch 1 Data Comparison

$UCL_{manual} = 12.61008 \qquad LCL_{manual} = 12.54432$



$UCL_{vision} = 12.66035 \qquad LCL_{vision} = 12.49479$

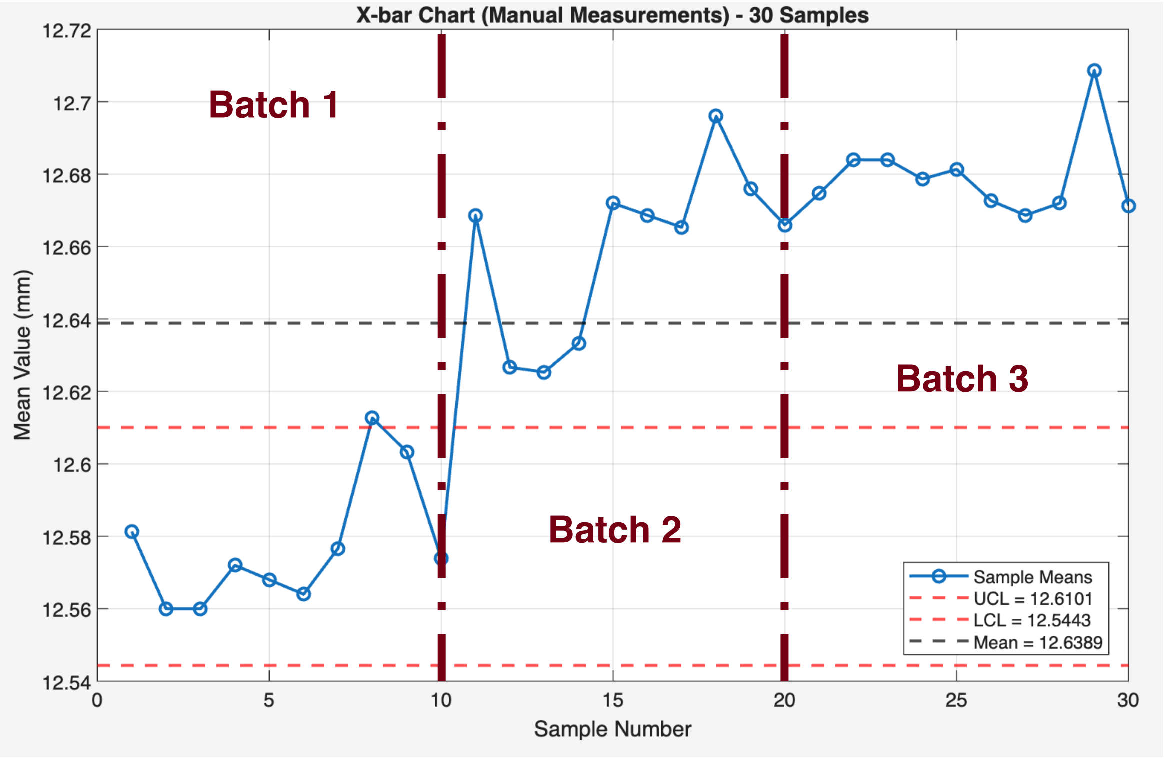


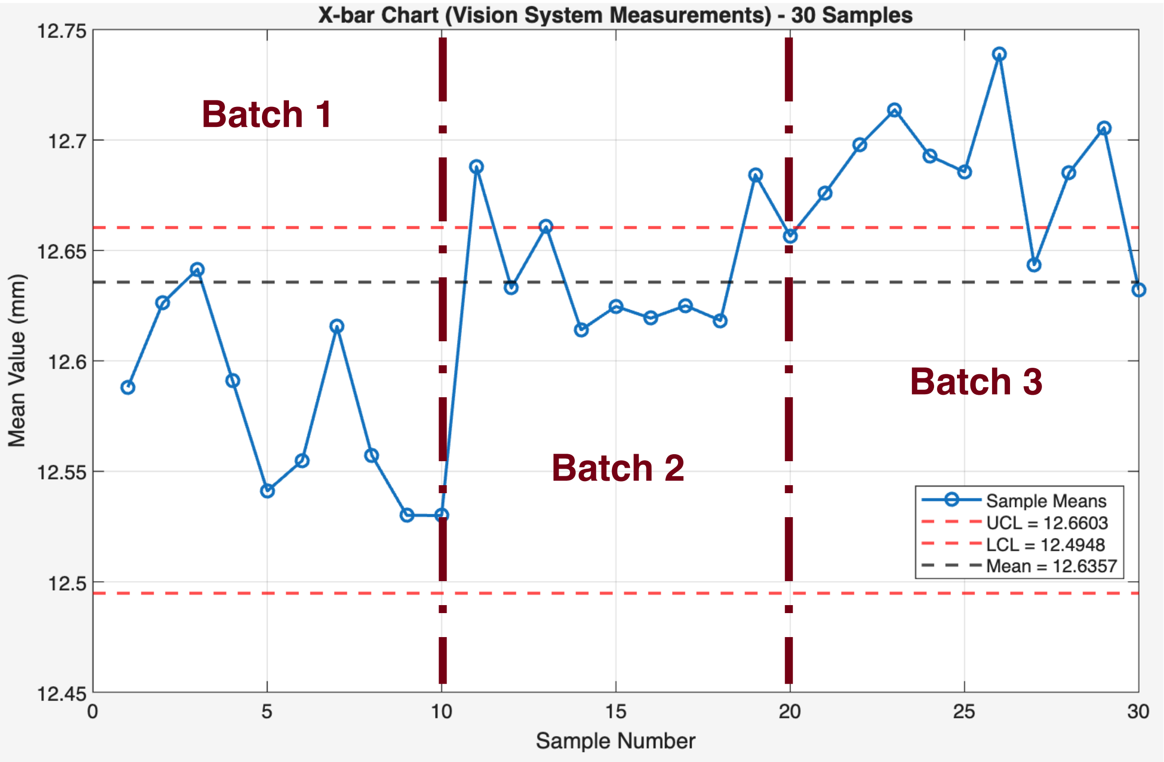
Since the group mean is within the control limits for both measurement types, it indicates that the Phase 1 data is within control.

Phase II Monitoring - Is it out of control? Variability?

This data represents batch 2 & 3 subgroups against batch 1 control limits.

Null Hypothesis: The mean shift between batches is not significant.





Phase II Monitoring

Mean Significance Test between Batches 1 & 2/3

$$t_1 = \frac{\bar{Y}_1 - \bar{Y}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{12.5776 - 12.6767}{\sqrt{\frac{0.0016652^2}{10} + \frac{0.0015145^2}{20}}} = 6.366299$$

$$By\ Welch's\ t - test\ DOF = 19$$

$$t_{crit} = 2.10$$

$$Therefore, \quad t_1 > t_{crit}$$

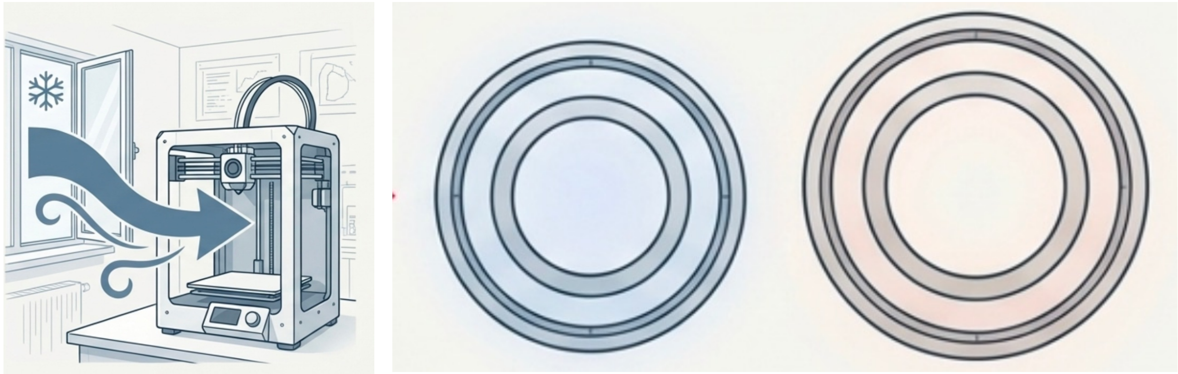
As a result, we reject the null hypothesis!

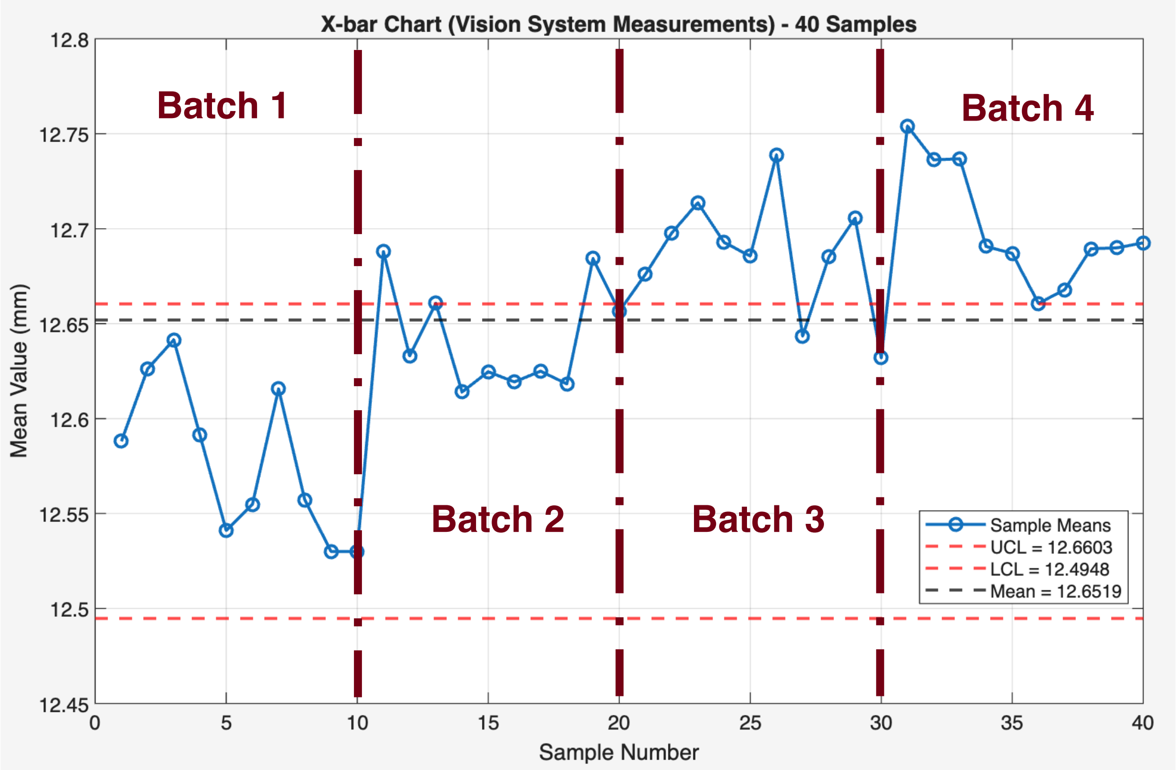
Mean Significance Test between Batches 2/3 & 4

$$t_2 = 0.554533 \qquad t_{crit} = 2.23$$
$$By\ Welch's\ t - test\ DOF = 10$$

$$Therefore, \quad t_2 < t_{crit}$$

As a result, we fail to reject the null hypothesis!

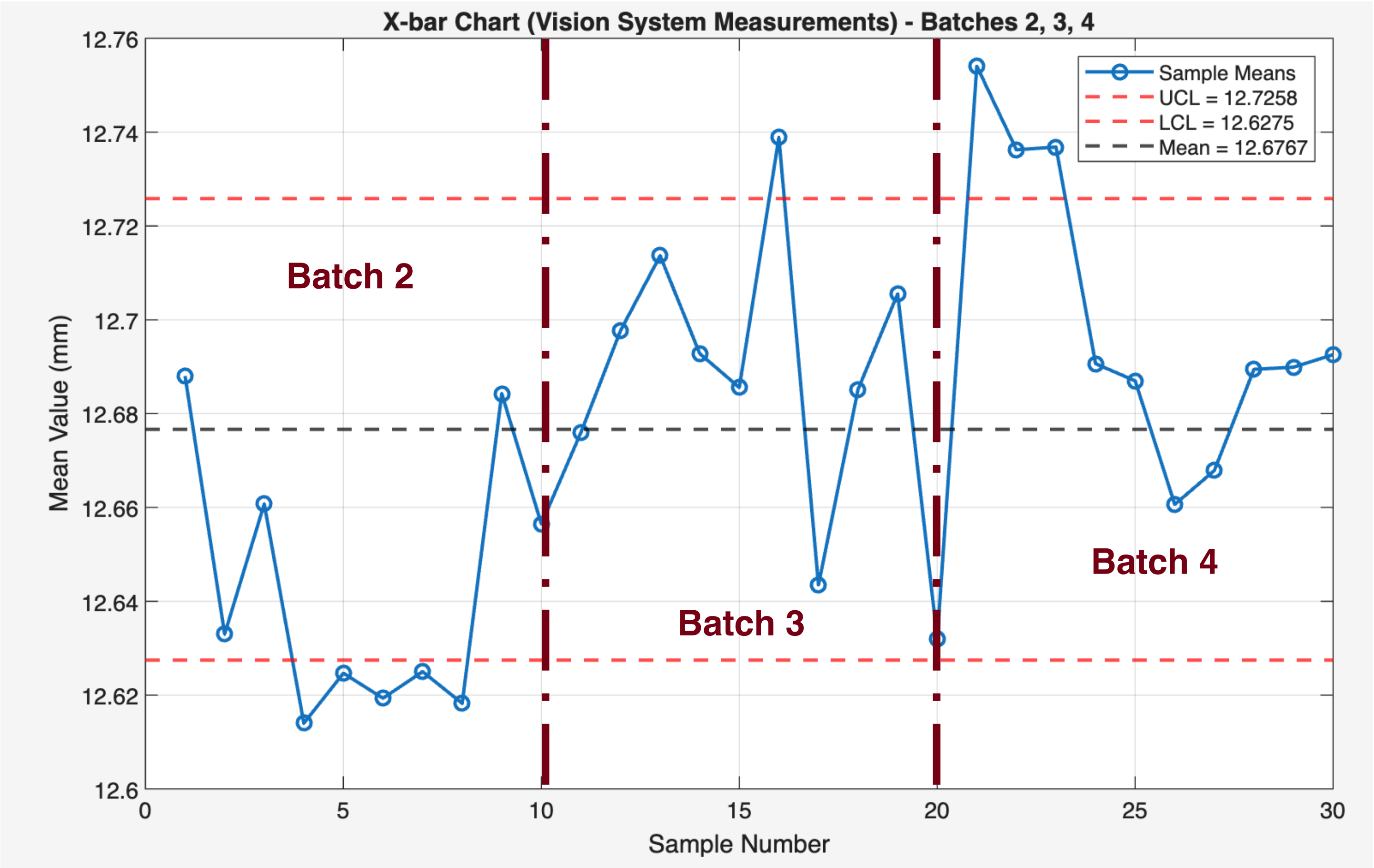




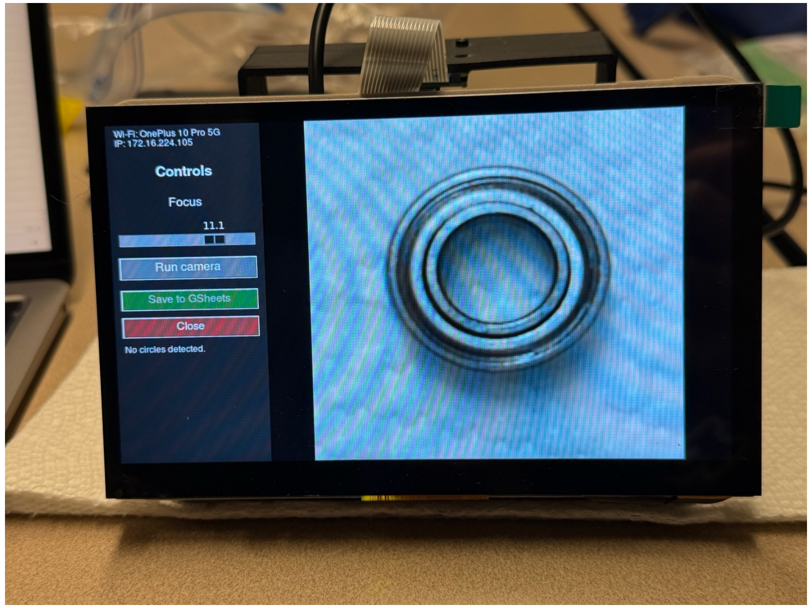
Final Control Chart

Since we establish that our Phase I data was influenced by an external deterministic effect, rebuilding the control chart using batches 2, 3, and 4 shows the process to be in control.

$UCL_{vision} = 12.7258$ $\quad LCL_{vision} = 12.6275$

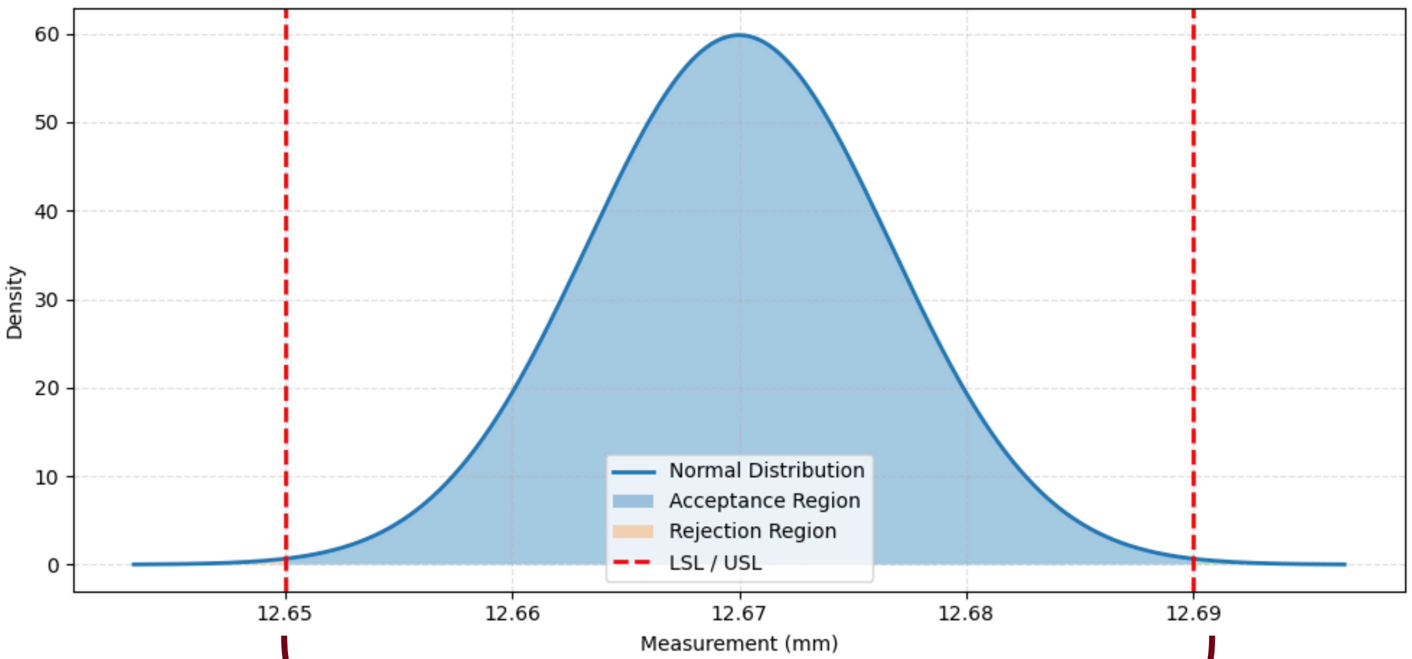


Identifying Design Specification Limits



LSL = 12.65

USL = 12.69



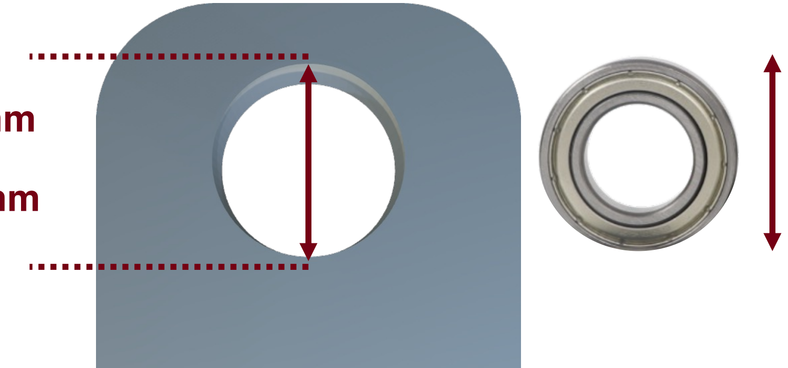
Key Parameters for Desired Bearing Press Fit:

- Average Bearing Diameter: 12.7 mm
- Variance of Bearing: 0.00762 mm (Negligible)
- Normal Fit Interference: 0.05 mm
- Loose Fit Interference: 0.01 mm
- Bambu A1 Shrinkage: ~0.1 to ~0.2mm (5% Variance)

99.7% Conforming Parts Desired

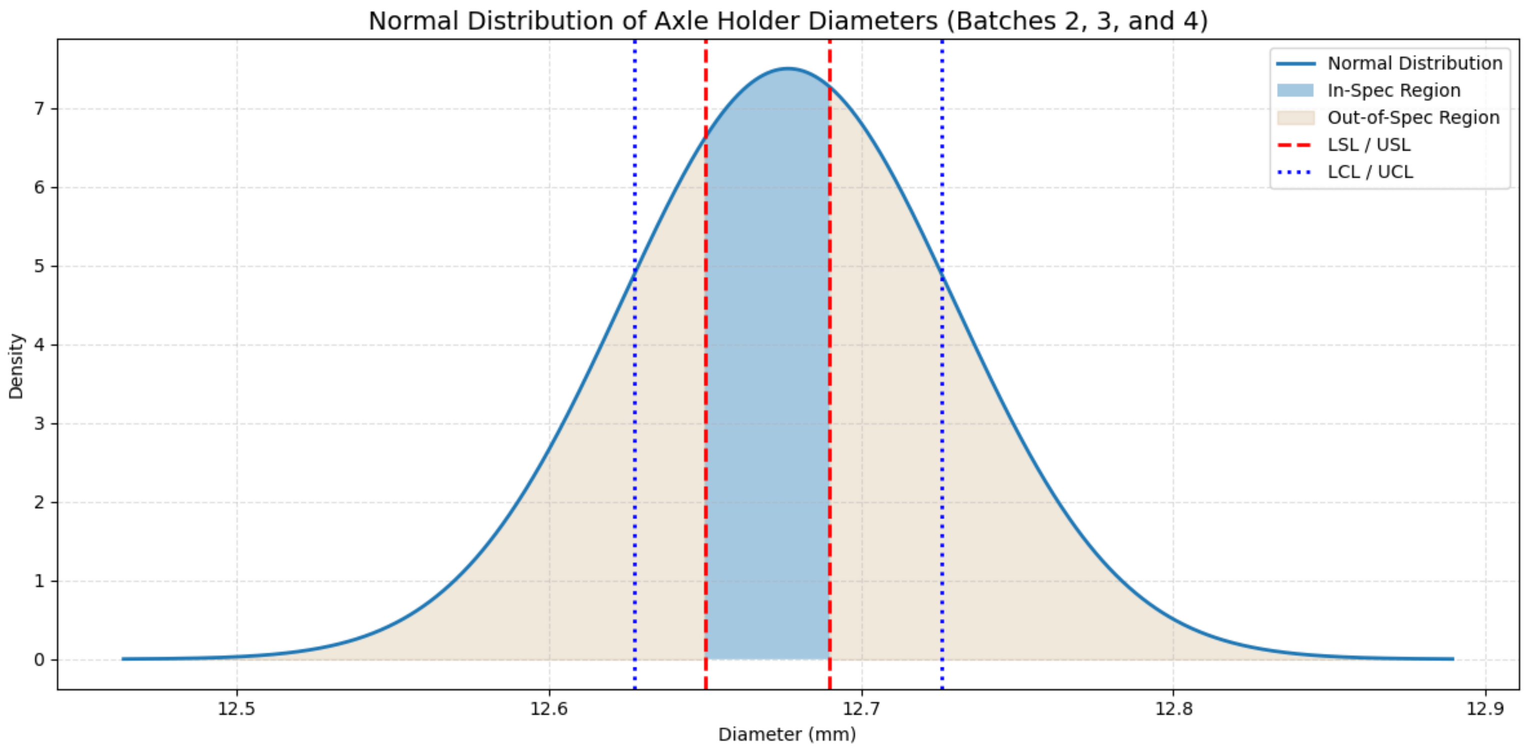
Min: 12.65 mm

Max: 12.69 mm



12.7 mm

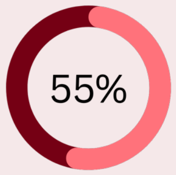
Comparison of Design Specification and Control Limits



Process is Centered But Control Limits Fall Outside Design Specification Region

Key Statistics

	Control Limits (mm)	Design Specification Limits (mm)
Lower Bound	12.6275	12.65
Upper Bound	12.7258	12.69



Parts Out of Specification
Prior to Optimization

$$c_p = 0.124$$

$$c_{pk} = 0.091$$

Design of Experiments – Identifying Relevant Factors

Factor	Low Level (-1)	High Level (1)
XY Speed (Nozzle Acceleration) (A)	6000 mm/s^2	8000 mm/s^2
Resolution (B)	0.012 mm	0.02 mm
Filament Material (C)	PLA	PETG
Cooling Fan (D)	Fan ON	Fan OFF

XY Speed (Nozzle Acceleration)

- Reasoning: This parameter changes how the nozzle changes speed. Higher acceleration value can lead to jerky movement, causing dimensional overshoot
    - The range available is 1,000 mm/s^2 to 10,000 mm/s^2
    - 6000 mm/s^2 and 8000 mm/s^2 is recommended by Bambu and provide reasonable print times.

Resolution

- Reasoning: Resolution affects the number of curves along a circular profile. Decreasing resolution increases curves, increases the clarity of the part.
    - The values selected are within the recommended range.

Filament Material

- Reasoning: Different materials have varying material properties.
    - PLA and PETG are readily available and have acceptable material properties for the application of these parts.

Cooling Fan

- Reasoning: An external cooling fan controls how fast deposited material cools, which influences dimensional stability
    - The external fan was turned either ON or OFF depending on the test that was being run

Design of Experiments – Fractional Factorial Design

Fractional Factorial Experiment

Run	A	B	C	D
1	-1	-1	-1	-1
2	+1	-1	-1	+1
3	-1	+1	-1	+1
4	+1	+1	-1	-1
5	-1	-1	+1	+1
6	+1	-1	+1	-1
7	-1	+1	+1	-1
8	+1	+1	+1	+1

Experimental Setup

- Full factorial experiment = 4 factor experiment = 2^4 = 16 runs required.
- Each run has 5 replicates, which take at least 45 min. to print.
- To reduce the print time required to complete all experiments, we design a fractional factorial experiment based on I = ABCD
  - This creates a balanced and orthogonal experiment

Effect	Aliased With
A	BCD
B	ACD
C	ABD
D	ABC
AB	CD
AC	BD
AD	BC

- Fractional Factorial Experiment = 2^{4-1} = 8 runs required
- The data generated through this experiment will be used to find the coefficients of this reduced linear regression equation:

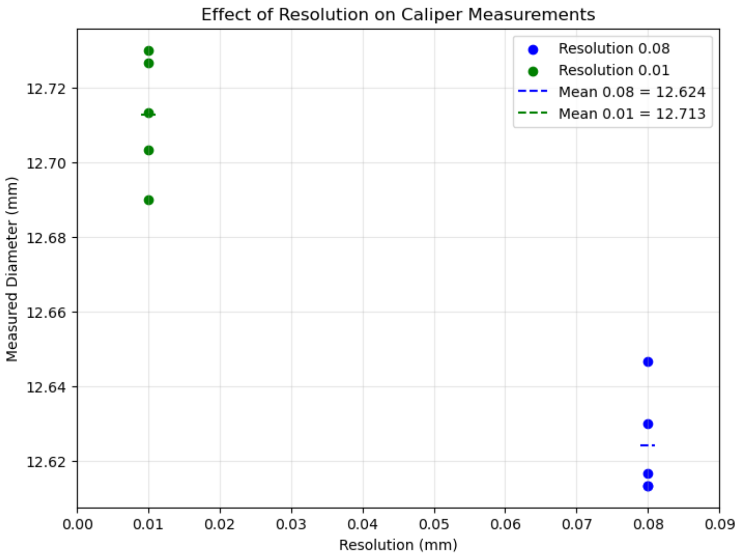
$$y = \beta_0 + \beta_1 A + \beta_2 B + \beta_3 C + \beta_4 D + \beta_5 AB + \beta_6 AC + \beta_7 AD$$

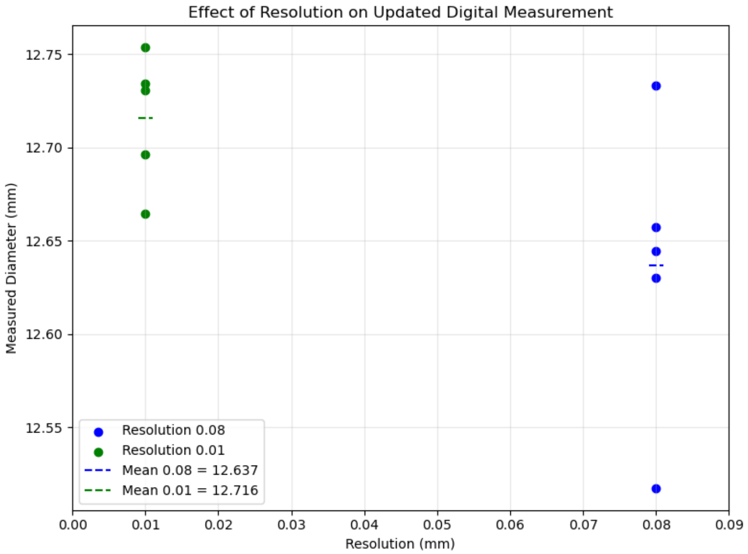
Design of Experiments – Regression Results

Manual Measurements	β_0	XY Speed β_A	Resolution β_B	Material β_C	Airflow β_D	β_{AB}	β_{AC}	β_{AD}	$F_{crit,95\%}$
Effects	12.629	0.0118	-0.012	-0.009	0.011	0.002	0.012	-0.010	
F-value Significance	-	14.28	9.17	9.33	13.67	0.53	14.06	10.69	4.15
Significant at 95%?	-	✓	✓	✓	✓	✗	✓	✓	

Vision System Measurements	β_0	XY Speed β_A	Resolution β_B	Material β_C	Airflow β_D	β_{AB}	β_{AC}	β_{AD}	$F_{crit,95\%}$	$F_{crit,90\%}$
Effects	12.539	-0.004	-0.003	-0.009	0.004	0.005	-0.006	-0.0006		
F-value Significance	-	3.15	1.87	70.16	3.78	4.48	6.14	0.07	4.15	2.87
Significant at 95%?	-	✗	✗	✓	✗	✓	✓	✗		
Significant at 90%?	-	✓	✗	✓	✓	✓	✓	✗		

Is Resolution Truly Significant?





Definition: In ***Bambu Studio***, resolution controls the maximum deviation allowed when simplifying model contours into arcs.

- **Lower resolution = higher accuracy** (more arcs, longer slicing time)
- **Higher resolution = lower accuracy**, which can distort geometry or cause layer misalignment
- Best used to balance detail preservation with efficient slicing.

Welch's t-test on Caliper Measurements:

$$t = \frac{\overline{Y}_1 - \overline{Y}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{12.713 - 12.624}{\sqrt{\frac{0.0148^2}{5} + \frac{0.0129^2}{5}}} = 10.136$$

$$By\ Welch's\ t-test\ DOF = 8$$

$$t_{crit,95\%} = 2.31$$

$$Therefore, \quad t > t_{crit}$$

SIGNIFICANT AT 95%

Welch's t-test on Vision System Measurements:

$$t = \frac{\overline{Y}_1 - \overline{Y}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{12.716 - 12.637}{\sqrt{\frac{0.0694^2}{5} + \frac{0.0316^2}{5}}} = 2.317$$

$$By\ Welch's\ t-test\ DOF = 6$$

$$t_{crit,95\%} = 2.45$$

$$Therefore, \quad t > t_{crit}$$

SIGNIFICANT AT 90%

DOE – Model Selection

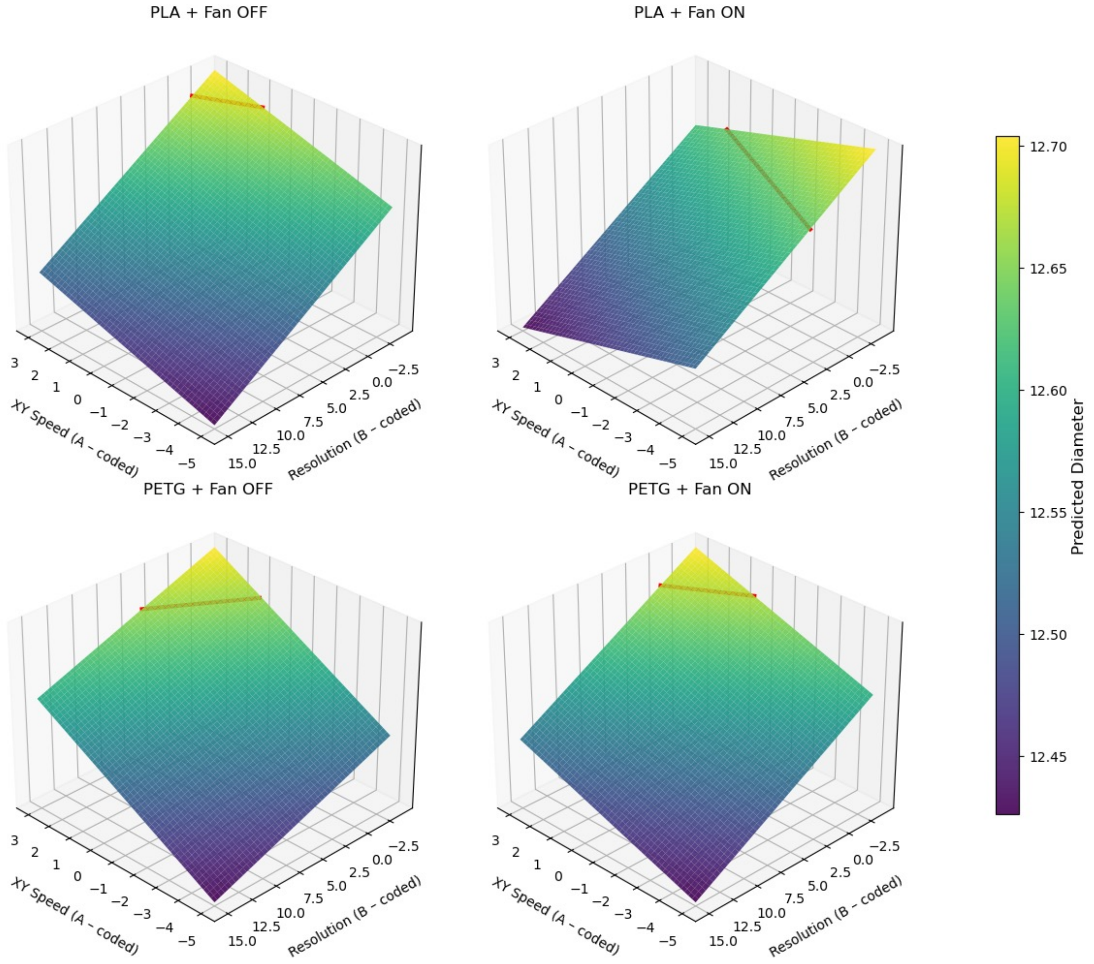
Measurement Type	XY Speed (A)	Resolution (B)	Material (C)	Airflow (D)	XY Speed – Material (AC)	XY Speed – Airflow (AD)
Vision System	✓	✗	✓	✓	✓	✓
Manual Measurement	✓	✓	✓	✓	✓	✓

Selected Regression Model

$$\boxed{y = 12.629 + 0.0118A - 0.00943B - 0.0115C - 0.0115D + 0.0117AC - 0.0102AD}$$

Optimizing for Parameter Selection

Response Surface



Objective

- Achieve a target **mean diameter of 12.67 mm**
- Print for the lowest **cost**, highest **speed**, lowest **variance**.

Final Choices

- **Filament:** PLA (Cut costs by $8 per kg of spool)
- **Fan:** Off (Reduce variables during print)

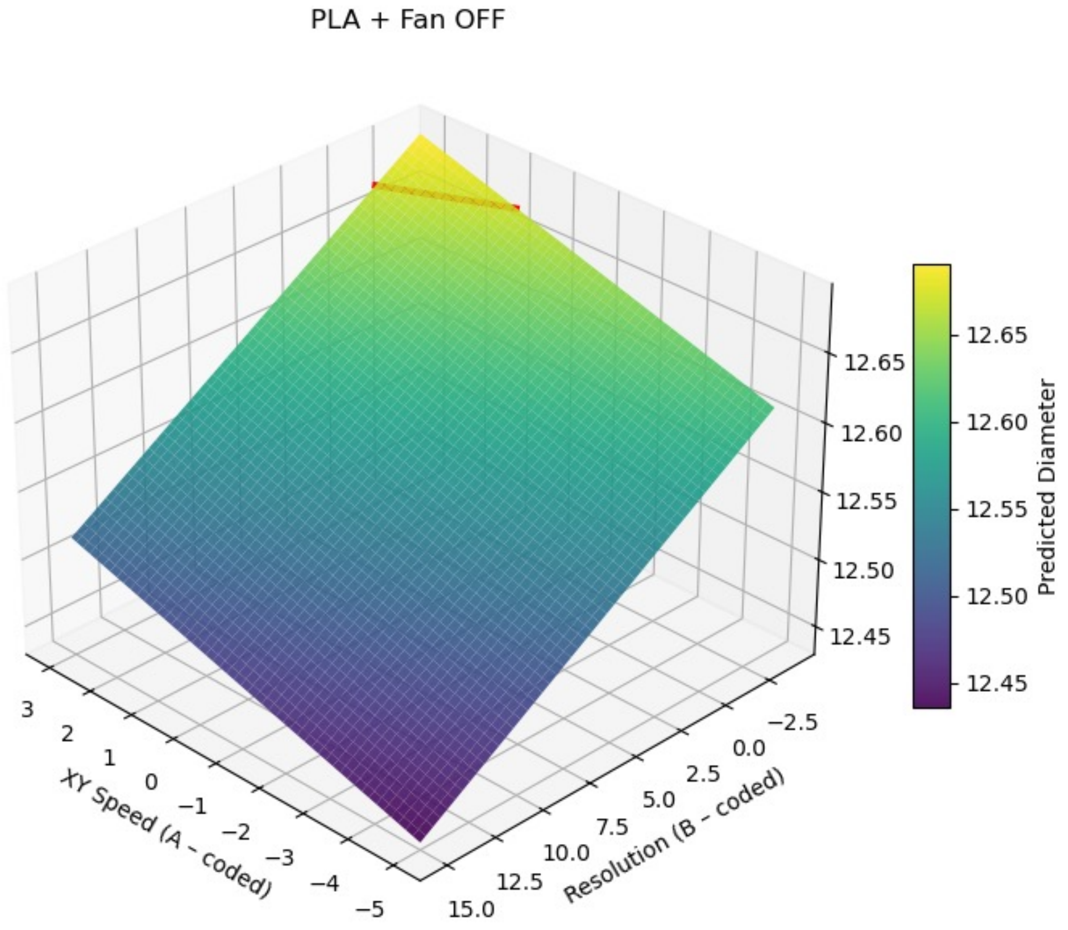
Sweep Through Continuous Parameters

- XY Speed (Acceleration)
- Resolution

These surfaces identify parameter regions that achieve the target mean diameter.

Optimizing for Parameter Selection

Response Surface



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Sweep Through Continuous Parameters

- **XY Speed** (Acceleration)
- **Resolution**

These surfaces identify parameter regions that achieve the target mean diameter.

Optimized Final Parameters

Parameters Chosen

Factor	Value
XY Speed (Nozzle Acceleration) (A)	10,000 mm/s^2
Resolution (B)	0.11 mm
Filament Material (C)	PLA
Airflow (D)	Fan OFF

Old Parameters

- **Total time** taken to print 1 RC Car: 58 min 17 sec
- **Parts Out of Spec**: 58.67%
- Old Material: PLA

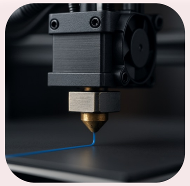
New Parameters (Printed 10 Samples)

- **Total time** taken to print 1 RC Car: 55 min 17 sec
- **Parts Out of Spec** (in 10 Samples): 20%
- New Material: PLA

Changes in Parameters

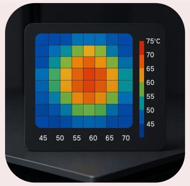
- Faster printing time by **3 minutes** (2-hour print reduction for 40 RC cars)
- Reduction in failure rate by **~28%** (Based on a small sample size)

Future Work



Expand Parameters by changing Nozzle Size

- Evaluating nozzle diameters (0.2 mm and 0.8 mm) to understand the trade-off between printing speed and dimensional accuracy.



Print-Bed Temperature Variance

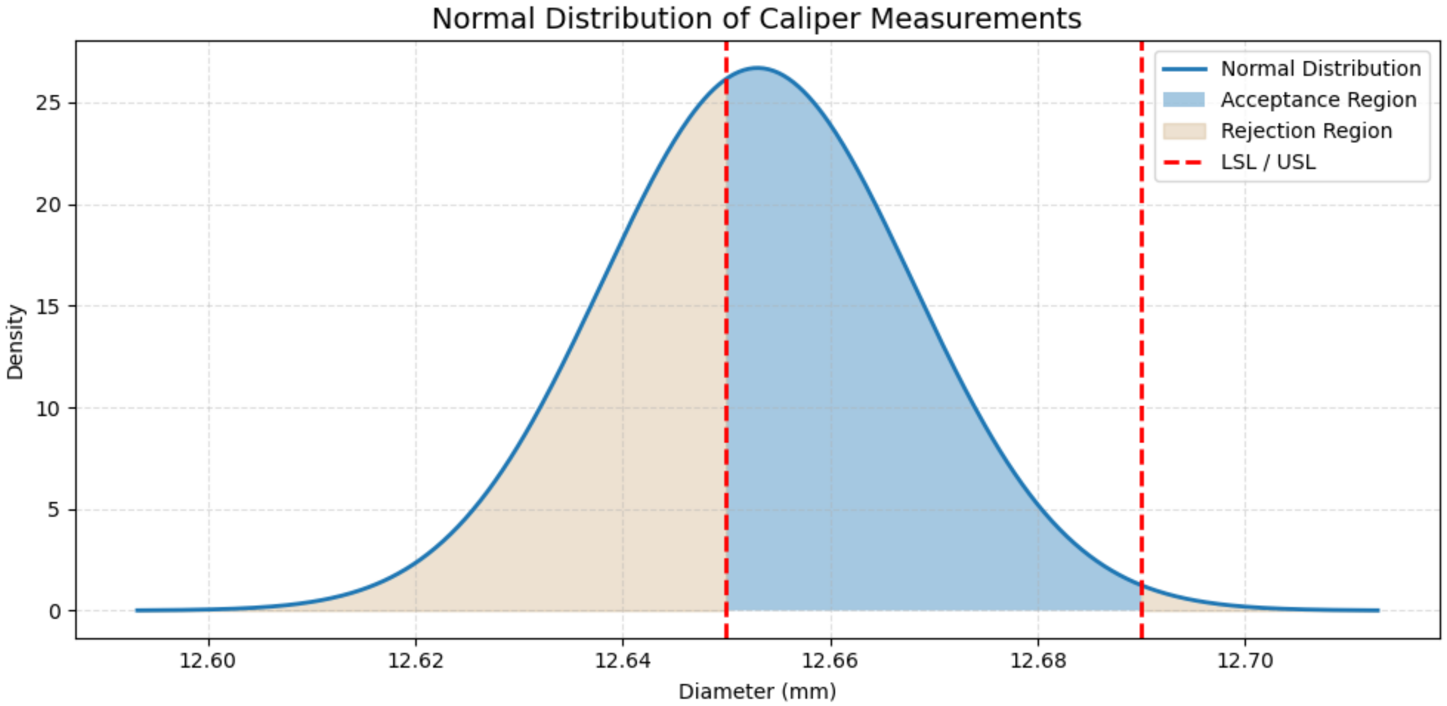
- Map dimensional variation across the entire print bed to detect temperature-driven regional effects.
- Develop correction factors or spatial compensation maps if systematic variation is found.

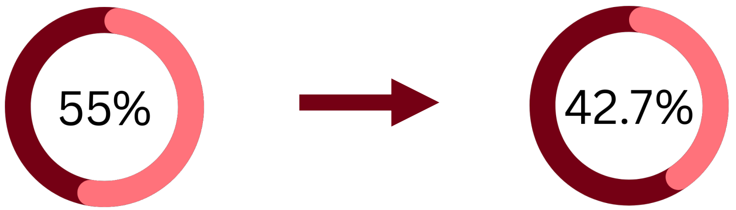
Improve Raspberry Pi Measurement System

- Enhance the vision system with controlled lighting, calibration artifacts, and refined edge-detection algorithms.
- Conduct a full Gage R&R study to quantify repeatability and reproducibility and guide hardware/software refinements.

Could we use these parameters to print other geometries?
Does variance and mean diameter trends hold across different feature shapes such as **slots, bushings, counterbores**.

Final Results & Conclusion





Parts Out of Specification
Prior to Optimization

Parts Out of Specification
Post Optimization

Takeaways and Analysis

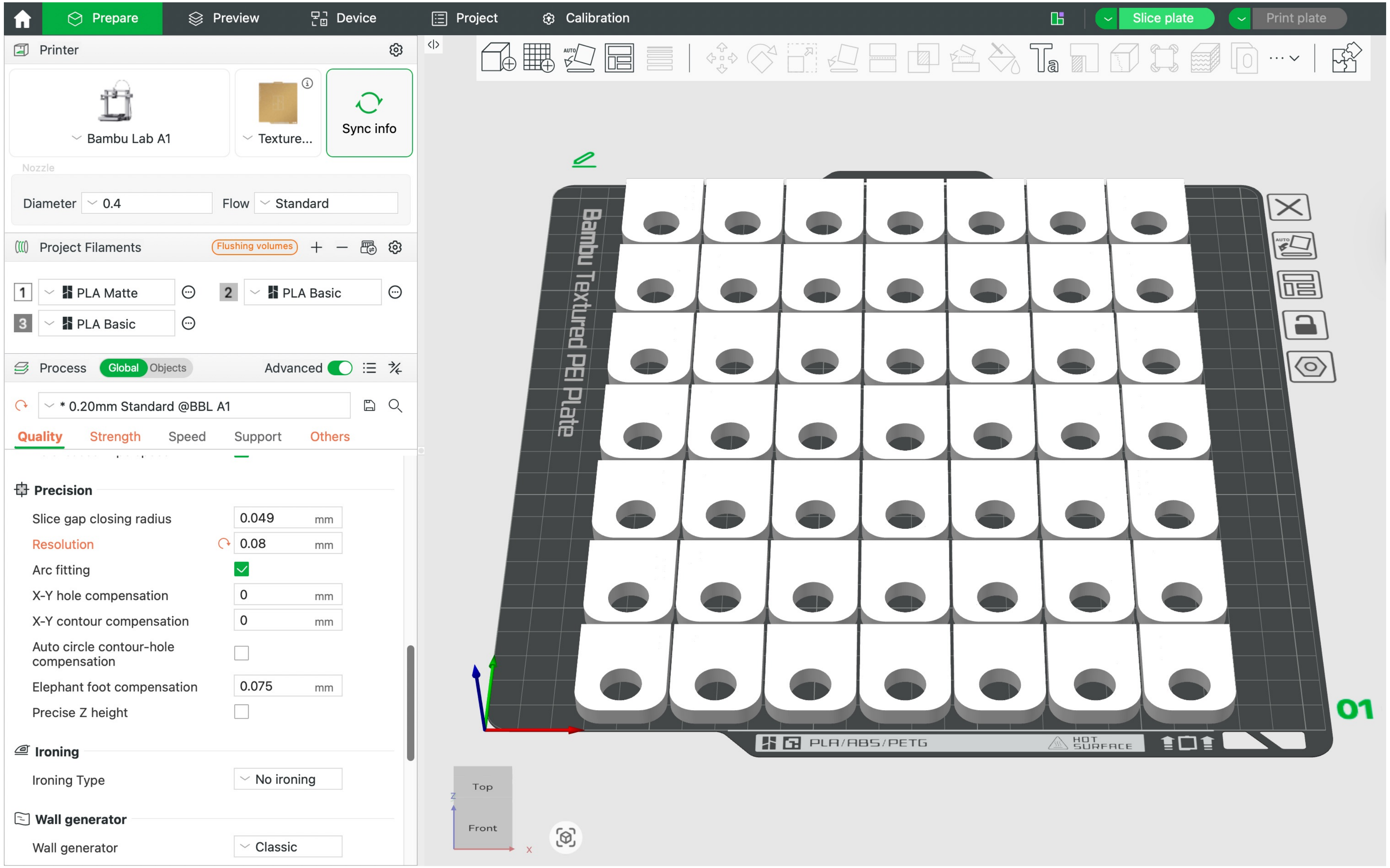
- Dimensional verification performed using caliper measurements.
  - **Mean diameter:** 12.653 mm
  - **Standard deviation:** 0.01494 mm
- Statistical model results:
  - Normal distribution predicted approximately **42% non-conforming parts.**
  - Reduced **failure rate from 58.67% to 42%** within a limited experimental window.

Reference

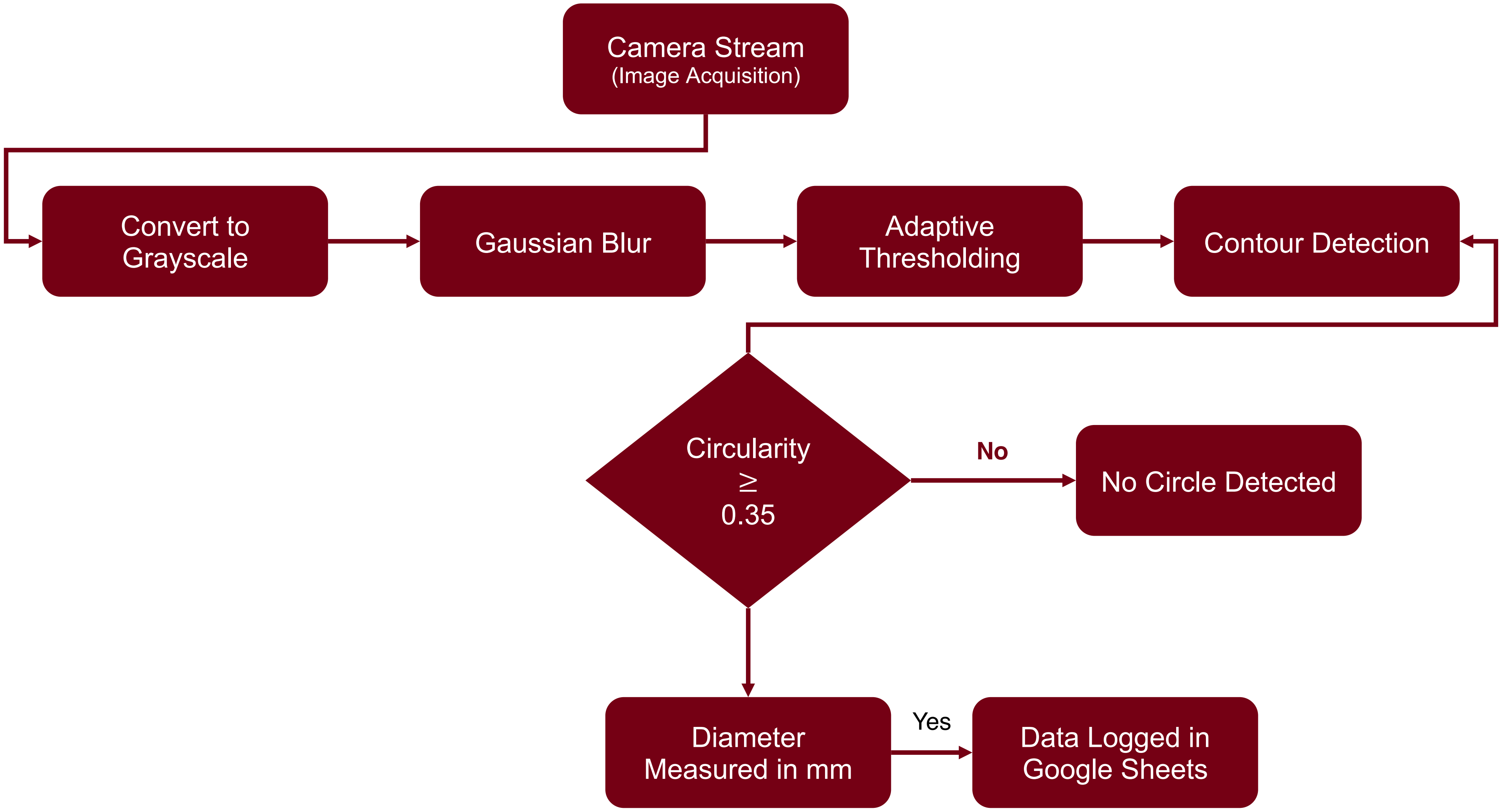
"3D Printer Accuracy, Tolerances and Engineering Fits Between 3D Printed Parts." *3D Print Files*, 3d-print-files.com, https://3d-print-files.com/3d-printer-accuracy-tolerances-engineering-fit-between-3d-printed-parts/. Accessed 1 Dec. 2025.

KMS Bearings, Inc. "Plastic Ball Bearings — Tolerances for Mounting and Interference Fits." *KMS Bearings — Technical Toolbox*, kmsbearings.com, https://kmsbearings.com/technical-toolbox/engineering-data/plastic-ball-bearings/. Accessed 3 Dec. 2025.

Appendix A – Printbed layout in slicer



Appendix B – Block Code



DOE – Model Selection

- The t-test using the manual data proved that the mean shift between a high-resolution value and low-resolution value is significant at a 95% confidence level

- While the t-test on the vision system data resulted in mean shift being insignificant at 95% confidence level, it only was so by a small margin.

- Shifting to a 90% margin causes the vision system data to be significant.

- Since the manual data indicates a strong significance at 95%, and the vision system data is significant at 90%, and only slightly insignificant at 95%. We can conclude that the mean shift caused by changing the resolution is significant.

- Therefore, we conclude that the model produced using the manual data is correct.

Selected Regression Model

$$\boxed{y = 12.629 + 0.0118A - 0.00943B - 0.0115C - 0.0115D + 0.0117AC - 0.0102AD}$$